Commuting Cost, Labor Market Power and Misallocation: Evidence from Chinese Bridge

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Abstract

Bridge

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1 Introduction

2 Background

3 Empirical Evidence

This section describes the reduced-form specifications that discipline the empirical facts and the quantitative model. The empirical exercise compares firms that receive larger commuting-time improvements from the cross-bay transport shock with firms that receive smaller improvements, before and after the opening of the bridge/tunnel system. The unit of observation is firm j in year t. The main outcomes are log employment, log wage, and log markdown:

$$y_{jt} \in \left\{ \log L_{jt}, \log w_{jt}, \log \nu_{jt} \right\}.$$

Employment and wages are observed in the firm panel. The markdown outcome is constructed from the production-function estimation and is used as the empirical counterpart to the labor-market-power object in the model below.

The treatment variable is built from the change in commuting times between residence locations and firms. Let d_{zj}^0 denote the travel time from residence location z to firm j in the no-bridge network, and let d_{zj}^1 denote the corresponding travel time in the network with the bridge/tunnel links. Let λ_z be the normalized working-age population share of location z, with $\sum_z \lambda_z = 1$. Firm j's market-access gain is

$$\Delta MA_j = \sum_z \lambda_z \left(d_{zj}^0 - d_{zj}^1 \right). \tag{1}$$

A positive value means that the cross-bay links reduce the population-weighted commuting time to the firm. The main treatment indicator is

$$BigMA_j = \mathbb{1}\left\{ \Delta MA_j \geq 0.5 \right\}, \tag{2}$$

where the threshold is measured in minutes. The post period is defined as $Post_t = \mathbb{1}\{t \geq 2011\}$.

The baseline difference-in-differences specification is

$$y_{jt} = \beta_1 BigMA_j \times Post_t + \mathbf{X}_{jt}' \Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}. \tag{3}$$

Firm fixed effects α_j absorb time-invariant firm characteristics and the level of the treatment variable. Year fixed effects λ_t absorb aggregate shocks common to all firms. The control vector $\mathbf{X}_{jt}$ includes firm age and year-varying controls for export status, ownership type, and two-digit industry. Standard errors are clustered at the town-year level, matching the geographic aggregation of the commuting shock and allowing arbitrary correlation among firms located in the same town and year.

To examine whether the transport shock reallocates workers differently across firms with different initial wage levels, the main heterogeneous specification interacts treatment with the firm's pre-shock wage. Let

$$\widetilde{w}_{j0} = \log w_{j0} - \overline{\log w_0},$$

where w_{j0} is firm j's initial wage and the mean is recomputed within the estimation sample. The heterogeneous difference-in-differences regression is

$$\begin{aligned} y_{jt} = {} & \beta_1 BigMA_j \times Post_t + \beta_2 \widetilde{w}_{j0} \times BigMA_j \times Post_t \\ & + \psi_t \widetilde{w}_{j0} + \mathbf{X}'_{jt}\Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}. \end{aligned} \tag{4}$$

The year-specific slopes $\psi_t \widetilde{w}_{j0}$ allow firms with different initial wages to follow different aggregate time paths even in the absence of differential exposure to the commuting shock. In this specification, β_1 measures the average post-opening effect for treated firms at the mean initial wage, while β_2 measures how this effect changes with the firm's initial wage. The employment version of equation (4) supplies two target moments used in the quantitative calibration. The wage version of the reduced-form specification supplies a third moment, the average wage response of firms with large market-access gains.

We also estimate event-study versions of equations (3) and (4). The baseline event-study equation is

$$y_{jt} = \sum_{\tau \neq \tau_0} \beta_\tau BigMA_j \times \mathbb{1}\{t = \tau\} + \mathbf{X}'_{jt}\Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}, \tag{5}$$

where τ_0 is a pre-opening reference year. The heterogeneous event-study additionally includes $\sum_{\tau \neq \tau_0} \gamma_\tau \widetilde{w}_{j0} \times BigMA_j \times \mathbb{1}\{t = \tau\}$ and year-specific initial-wage slopes. These coefficients are used to inspect pre-trends and to show how the post-opening effect evolves around the shock.

Table 1: DID Regression Results

	(1) lnemp	(2) lnemp	(3) lnmd	(4) lnmd	(5) lnw	(6) lnw
$BIG \times post$	-0.012	-0.382***	-0.056**	-0.428**	0.065***	0.285*
	(0.025)	(0.135)	(0.023)	(0.178)	(0.022)	(0.170)
$BIG \times post \times \log w_0$		0.094**		0.114**		-0.064
		(0.042)		(0.054)		(0.053)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,350	16,411	5,066	5,062	20,025	16,145

Notes: Regressions are estimated using `reghdfe`. Standard errors are clustered at the town-sector level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4 Model Setup

This section presents the quantitative spatial model used to interpret the reduced-form evidence. Workers live in residence locations $z \in Z$, interpreted as streets or towns, and firms are indexed by $j \in J$. Each firm belongs to a sector $s(j) \in S$, and J_s denotes the set of firms in sector s. The mass of workers living in z is l_z, normalized so that $\sum_z l_z = 1$. Commuting time from z to j is d_{zj}. A firm is characterized by a wage w_j, a productivity term A_j, and a firm amenity a_j. The amenity captures non-pecuniary firm attributes that affect worker choices but do not enter the production function directly.

4.1 Worker Choice

Worker i, living in location z, obtains indirect utility from working at firm j,

$$U_{izj} = \log w_j + a_j - \eta \log d_{zj} + \frac{1}{\theta}\xi_{ij}, \tag{6}$$

where $\eta > 0$ governs the disutility of commuting time and $\theta > 0$ governs the sensitivity of worker choices to systematic utility. The vector of idiosyncratic taste shocks follows a nested multivariate extreme-value distribution:

$$F(\xi_{i1}, \ldots, \xi_{iJ}) = \exp\left\{ -\sum_{s\in S} \left[\sum_{k\in J_s} \exp\left(-\frac{\xi_{ik}}{\sigma} \right) \right]^{\sigma} \right\}, \qquad 0 < \sigma \leq 1. \tag{7}$$

The nest index s denotes sector. This structure allows idiosyncratic preferences over firms in the same sector to be correlated, while $\sigma = 1$ collapses the model to the one-level logit case.

Let

$$Q_{zj} = w_j \exp(a_j) d_{zj}^{-\eta}$$

denote the systematic attractiveness of firm j to workers living in z. The probability that a worker in z chooses sector s is

$$\pi_{zs} = \frac{\left[\sum_{k \in J_s} Q_{zk}^{\theta/\sigma}\right]^{\sigma}}{\sum_{r \in S} \left[\sum_{k \in J_r} Q_{zk}^{\theta/\sigma}\right]^{\sigma}}. \tag{8}$$

Conditional on choosing sector s, the probability of choosing firm $j \in J_s$ is

$$\pi_{zj|s} = \frac{Q_{zj}^{\theta/\sigma}}{\sum_{k \in J_s} Q_{zk}^{\theta/\sigma}}. \tag{9}$$

The unconditional probability of choosing firm $j \in J_{s(j)}$ is

$$\pi_{zj} = \pi_{z,s(j)} \pi_{zj|s(j)}. \tag{10}$$

Firm j's labor force is the sum of the workers choosing that firm across residence locations:

$$L_j = \sum_{z \in Z} l_z \pi_{zj}. \tag{11}$$

The firm-specific labor supply elasticity follows from the nested logit choice structure. For workers in residence location z, the elasticity of the choice probability with respect to firm j's own wage is

$$\varepsilon_{zj} = \frac{\partial \log \pi_{zj}}{\partial \log w_j} = \theta \left[\frac{1}{\sigma} + \left(1 - \frac{1}{\sigma}\right) \pi_{zj|s(j)} - \pi_{zj}\right]. \tag{12}$$

When $\sigma = 1$, equation (12) becomes the one-level-logit expression $\varepsilon_{zj} = \theta(1 - \pi_{zj})$. Let

$$\omega_{zj} = \frac{l_z \pi_{zj}}{L_j}$$

be the share of firm j's workforce that comes from residence location z. Firm j's aggregate labor supply elasticity is the workforce-share-weighted average of the origin-specific elasticities:

$$\varepsilon_j = \frac{\partial \log L_j}{\partial \log w_j} = \sum_{z \in Z} \omega_{zj} \varepsilon_{zj}. \tag{13}$$

The bridge changes commuting times, and hence changes sector choices π_{zs}, within-sector firm choices $\pi_{zj|s}$, unconditional firm choices π_{zj}, worker-origin composition ω_{zj}, and the labor supply elasticity ε_j.

4.2 Firms and Markdown

Firm j produces with decreasing returns to labor,

$$Y_j = A_j L_j^{\alpha}, \qquad 0 < \alpha < 1, \tag{14}$$

with product prices normalized. The firm chooses wages while recognizing that its labor supply is upward sloping. The first-order condition equates the marginal product of labor to marginal labor expenditure:

$$\alpha A_j L_j^{\alpha-1} = w_j \left(1 + \frac{1}{\varepsilon_j}\right). \tag{15}$$

Equivalently, the equilibrium wage satisfies

$$w_j = \alpha A_j L_j^{\alpha-1} \frac{\varepsilon_j}{1+\varepsilon_j}. \tag{16}$$

The model markdown is defined as the ratio between the marginal product of labor and the wage:

$$\nu_j = \frac{\alpha A_j L_j^{\alpha-1}}{w_j} = 1 + \frac{1}{\varepsilon_j} > 1. \tag{17}$$

Thus, a firm with a lower labor supply elasticity has greater labor-market power and a higher markdown.

4.3 Equilibrium, Inversion, and Bridge Counterfactual

Given primitives $\{l_z, d_{zj}, A_j, a_j, s(j)\}$ and parameters $(\alpha, \eta, \theta, \sigma)$, an equilibrium is a vector of wages $\{w_j\}_{j\in J}$ such that worker choices satisfy equations (8)–(10), firm labor satisfies equation (11), firm-level elasticities satisfy equation (13), and wages satisfy equation (16). Since only relative wages matter for worker choices, the numerical solution normalizes the aggregate wage bill.

For a candidate value of (η, θ, σ), the model maps the observed baseline economy into firm primitives in two steps. First, firm amenities are inverted from observed baseline employment and wages:

$$L_j^{data} = \sum_z l_z \pi_{zj}\left(w^{data}, d^0, a; \eta, \theta, \sigma\right), \qquad \frac{1}{J}\sum_j a_j = 0. \tag{18}$$

Second, firm productivity is backed out from the wage first-order condition:

$$A_j = \kappa \frac{w_j^{data}\left(1+\varepsilon_j^{data}\right)}{\alpha\left(L_j^{data}\right)^{\alpha-1}\varepsilon_j^{data}}, \tag{19}$$

where κ is a normalization constant. In the quantitative exercise, α is fixed and η, θ, and σ are calibrated to match the reduced-form moments described below.

The counterfactual compares two commuting networks. The baseline network uses no-bridge commuting times d_{zj}^0. The post-opening network uses with-bridge commuting times d_{zj}^1. Holding residence populations, firm amenities, and firm productivities fixed, we solve the model under d^0 and d^1. The model-implied changes in employment, wages, and markdowns are then

$$\Delta \log L_j = \log L_j^1 - \log L_j^0, \qquad \Delta \log w_j = \log w_j^1 - \log w_j^0, \qquad \Delta \log \nu_j = \log \nu_j^1 - \log \nu_j^0.$$

For the calibration moments, the model appends the no-bridge equilibrium and the with-bridge equilibrium as a two-period firm panel and estimates the analogue of equation (4) with outcome $\log L_j$, firm fixed effects, period fixed effects, and period-specific initial-wage slopes. The two model coefficients on $BigMA_j \times Post_t$ and $\widetilde{w}_{j0} \times BigMA_j \times Post_t$ are matched to their empirical counterparts. The model also estimates the corresponding two-period wage response, using outcome $\log w_j$, to match the reduced-form wage moment.

5 Calibration and Simulation

The calibration chooses the commuting-cost parameter η, the choice-sensitivity parameter θ, and the sector-nest parameter σ so that the model reproduces the reduced-form responses of employment and wages to the bridge-induced market-access shock. The production curvature α is fixed. For each candidate triple (η, θ, σ), the model first inverts the baseline firm amenities and firm productivities using equations (18) and (19). Holding these primitives fixed, the model then solves the no-bridge equilibrium under d^0 and the post-opening equilibrium under d^1. The resulting firm-level changes in employment and wages are converted into a two-period model panel and projected onto the same treatment and wage-heterogeneity variables used in the reduced-form regressions.

The target vector contains three reduced-form coefficients:

$$\widehat{m}^{RF} = \begin{pmatrix} \widehat{\beta}_{L,1}^{RF} \\ \widehat{\beta}_{L,2}^{RF} \\ \widehat{\beta}_{w,1}^{RF} \end{pmatrix}. \tag{20}$$

The first element, $\widehat{\beta}_{L,1}^{RF}$, is the coefficient on $BigMA_j \times Post_t$ in the employment regression. The second element, $\widehat{\beta}_{L,2}^{RF}$, is the coefficient on $\widetilde{w}_{j0} \times BigMA_j \times Post_t$ in the employment regression and captures whether the employment effect of the bridge differs between high- and low-wage firms. The third element, $\widehat{\beta}_{w,1}^{RF}$, is the coefficient on $BigMA_j \times Post_t$ in the wage regression. These three moments are computed from the reduced-form sample and then held fixed as calibration targets.

For any candidate (η, θ, σ), the model generates the corresponding moment vector

$$m^M(\eta, \theta, \sigma) = \begin{pmatrix} \beta_{L,1}^M(\eta, \theta, \sigma) \\ \beta_{L,2}^M(\eta, \theta, \sigma) \\ \beta_{w,1}^M(\eta, \theta, \sigma) \end{pmatrix}, \tag{21}$$

where $\beta_{L,1}^M$ and $\beta_{L,2}^M$ are obtained by applying the employment calibration regression to the model-generated two-period panel, and $\beta_{w,1}^M$ is obtained by applying the wage calibration regression to model-generated wages. The estimates $(\widehat{\eta}, \widehat{\theta}, \widehat{\sigma})$ solve

$$(\widehat{\eta}, \widehat{\theta}, \widehat{\sigma}) = \arg\min_{\eta,\theta,\sigma} \left[m^M(\eta, \theta, \sigma) - \widehat{m}^{RF}\right]' W \left[m^M(\eta, \theta, \sigma) - \widehat{m}^{RF}\right], \tag{22}$$

where the calibration uses an identity weighting matrix W and searches over a compact admissible parameter space. The two employment moments discipline the magnitude and wage heterogeneity of labor reallocation induced by the commuting-cost shock. The wage moment adds information about the equilibrium wage response, which is shaped by both labor reallocation and changes in firms' labor supply elasticities. Since commuting-time changes enter worker choice through the commuting-cost parameter, the overall choice sensitivity, and the sector-nest parameter, the wage and elasticity channels are important for separating the commuting-cost response from substitution across and within sectors.